ثانوية المدينة التعليمية
Education City High School

1.7

Rational Functions and End Behavior

Student Notes and Guided Practice

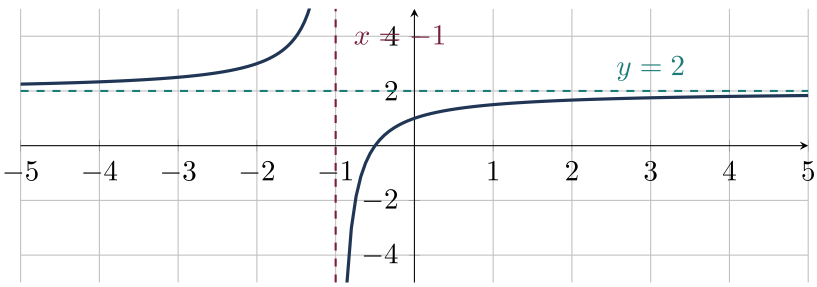
AP Precalculus - Unit 1 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Learning Targets

- Determine horizontal, slant, and higher-degree polynomial asymptotes.
- Use polynomial division to explain rational-function end behavior.
- Interpret long-run behavior in context.

Essential Question

How does the degree comparison in a rational function determine its end behavior?



Concept Framework

Key Idea

If numerator degree is less than denominator degree, the horizontal asymptote is $y = 0$.

Key Idea

Equal degrees give a horizontal asymptote equal to the ratio of leading coefficients.

Key Idea

If numerator degree is exactly one greater, polynomial division gives a slant asymptote.

Key Idea

The remainder term divided by the denominator approaches 0 when the denominator degree exceeds the remainder degree.

Formula Map

Formula and Structure

$$R(x) = \frac{P(x)}{Q(x)} = S(x) + \frac{A(x)}{Q(x)}$$

Formula and Structure

$$\deg A < \deg Q \Rightarrow R(x) - S(x) \to 0$$

Worked Examples

Worked Example 1

Problem. Find the horizontal asymptote of $\frac{5x^3-2}{2x^3+x}$.
Solution. Equal degrees; the asymptote is $y = 5/2$.

Worked Example 2

Problem. Find the slant asymptote of $\frac{x^2+3x+5}{x+1}$.
Solution. Division gives $x + 2 + 3/(x+1)$, so the slant asymptote is $y = x + 2$.

Worked Example 3

Problem. Why may a rational graph cross a horizontal asymptote?
Solution. An asymptote describes end behavior, not a barrier. The function can equal the asymptote at finite inputs.

Multiple-Representation Connection

AP Reasoning

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. Determine the end-behavior asymptote of $\frac{3x^2+1}{x^3-2}$.

2. Determine the end-behavior asymptote of $\frac{5x^3-x}{2x^3+7}$.

3. Determine the end-behavior asymptote of $\frac{x^2+4x+1}{x-1}$.

4. Determine the end-behavior asymptote of $\frac{2x^4+1}{x^2+3}$.

5. Determine the end-behavior asymptote of $\frac{-4x^5+x}{2x^5-1}$.

6. Divide $\frac{2x^3-3x^2+4}{x^2+1}$ and state the asymptote.

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is ______________________________.
2. A restriction I must remember is ______________________________.
3. One graphical feature I can justify algebraically is ______________________________.

Video support: Flipped Math topic page | Original ECHS material informed by Pearson and Holt Precalculus concepts.